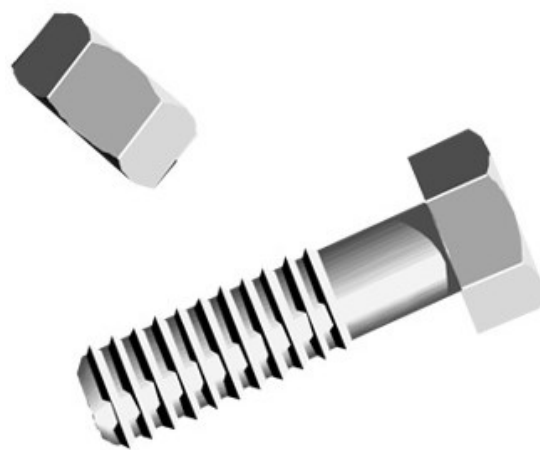


GE Vernova

-Original Document-

Technical Documentation Wind Turbine Generator Systems 1&2MW Platform



Bolt Torque Specification

Rev. AF - Doc-0000921 - EN 2025-02-13



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Document Revision Table

Rev.	Date (YYYY/MM/DD)	Affected Pages	Change Description
24	2022/12/13	36	UPDATED values for Tower flange / Yaw bearing
		41	UPDATED torque for 1MW – 77, 82.5, 87, 91, 100, 103 Rotor Hub to main shaft flange
		41	UPDATED row for All other 1.x variants Rotor Hub to main shaft flange
		41, 42, 43	UPDATED notes for 231mm Rotor Hub to main shaft flange
AD	2023/10/03	19	UPDATED section name 5.2.4.1.1
		47	UPDATED and COMBINED Table 6 and Table 7
		47	REMOVED LM47.3 from Table 5
		37	UPDATED Permit to Shrink Disc
AE	2024/01/22	20, 47	UPDATED Figure 9 and Table 6
			DELETED Section 'Specific Torque Check for Rotor Blade to Pitch Bearing with M36 Stud Type Fasteners and Wedgelocks'
AF	2025/02/13	19	ADDED Section 5.2.4.1
		21	UPDATED Section 5.2.4.3
		27	UPDATED Section 6.4.2
		31	UPDATED Section 6.5.2
		37, 38, 39, 42, 43,44	UPDATED Table 4
		47	UPDATED Table 6

1 Document Scope

Platform	1&2MW
Product¹	1.5-70.5, 77, 82.5 1.6-77, 82.5, 87, 91, 97, 100 1.7-100, 103, 107 1.85-82.5, 87 2.0 to 2.8- 107, 116, 127, 132 All RePower Wind Turbines based on the above original nameplates
Countries / Regions	ALL
WTG Specific Configuration	ALL

Table 1: Document Scope

2 Purpose of the Document

This document defines the torque or tensioning requirements (assembly, erection, and maintenance) for all fasteners used on the 1&2MW Platform wind turbine. This does not apply for customer provided components.

3 Acronyms

Term	Meaning
BIM	Break-in Maintenance
CCW	Counter-Clockwise
DT	Drive Train
EHS	Environment Health and Safety
SBI	Single Blade Installation
SCADA	Supervisory Control and Data Acquisition
WTGS	Wind Turbine Generator System

Table 2 : Acronyms

4 General

4.1 Restrictions and Warnings

Specifications/requirements may change at any time, check for updates prior to performing operations.

All hardware must be torqued or tensioned in accordance with this document unless otherwise specifically indicated by a Non-GE design or manufacturer requirement, or an update from GE. Assembly drawings or assembly specifications from other vendors may supersede this document. If in doubt consult GE Renewables.

Lubrication must not be used when the fasteners are attached to or near electronic equipment unless specified by the controlling drawing or assembly procedure.

No lubrication can be present on the mating surfaces of the bolt head, nuts, washers, etc. except as specified in Section 5.1 “General Installation Requirements for ALL Torqued Bolts”.

¹ Rated Power (MW) Rotor Diameter (m) Grid Frequency (Hz)

During WTG installation, additional requirements exist as outlined in the installation manual which include, but are not limited to, daily tool calibration checks. Consult the installation manual for additional information.

Due to the risk of injuries during torquing operations with hydraulic torque tools, the following guidance is given when using hydraulic torque devices:

1. **HANDLES TO BE USED** with all torque heads on applications where the Technician is required to make hand contact with the tool while the pump is running.
2. **SINGLE PERSON TORQUE OPERATION MUST BE FOLLOWED** on all fasteners, per manufacturer instruction for all torque operations. Exceptions are made for Pitch Bearing-to-Hub as it is understood that this may present challenges based on the configuration of the hydraulic torque pump remote.
- 2A. If two-person operation is required. The recommendation is to turn off the pump when positioning the tool head to avoid accidental actuation of the hydraulic tool. This practice gives an audible warning to the Technician when the tool is in a safe/unsafe condition.

NOTICE

Note for all level 1 components connections: Base Flange-To-Anchor bolt, Tower Flange-to-Tower Flange, Top Tower Flange-To-Yaw Bearing, Rotor-To-Hub Adaptor, Pitch Bearing-To-Hub & Rotor Blade-To-Pitch Bearing.

- ▶ Mixing of fastener hardware (bolt, washer & nuts) between manufacturer's is not authorized on each fastener location.
- ▶ Two different manufacturer's fasteners may be used on one joint with no more than 10 % of the fasteners being from the second manufacturer.

4.2 Re-use of Fasteners

It is NOT acceptable to re-use previously torqued or tensioned hardware for components listed in Table 4 unless specified within the re-use column or otherwise approved by GE Engineering. Unauthorized re-use of fasteners may cause them to fail.

Requirements for re-use of fasteners

If any condition is not met the hardware must be replaced.

- Joints must not be loose and no more than 10 % of fasteners fail maintenance torque check.
- No cracks, deformation (bending or stretching) evident anywhere on the fastener.
- No galling / deformation / damage on threads or washer.
- Nut must be able to be run by hand over the threads above/through/below the engaged area.

Fasteners listed as single re-use have the following additional requirements:

- No indication of prior re-use (as documented below)
- Required documentation:
 - Entry made in turbine log and written message on joint indicating re-use. If fasteners are replaced at later time, then line out message on joint with date and make entry into turbine log.
 - Stud ends are colored in with paint pen and if studs are exposed to weather the end must additionally be marked by hitting it with a punch to indicate re-use.

4.3 Tools

	FURTHER INFORMATION	
Only calibrated tool systems with a maximum torque tolerance of $\pm 6\%$ must be used! The contractor for projects & installation, will have additional requirements that outlined in the installation manual which include, but are not limited to, daily tool calibration checks. To check measuring and test equipment for damage and accuracy prior to each use. This is to ensure the measuring equipment has not suffered deterioration in accuracy, prior to its next calibration expiration date. Consult the installation manual for additional information. Calibration certificates are accepted from tool manufacture and/or 3rd party. Daily calibration checks shall be performed at the torque value in the BTS for the connection the tool will be utilized.		

NOTICE
Pressure, force, resistance, torque, distance and measuring tools shall be calibrated against standards traceable to the National Institute of Standards and Technology (NIST) measurement standards or ISO/IEC 17025 in accordance with ISO 10012 or ANSI and all where applicable.

	WARNING	
Risk of Serious Injury! ▶ POTENTIAL PINCH POINT - Only certified and competent persons to operate the tools. ▶ Always use the safety handle to keep hands free of the reaction arm		

5 Bolt Torque Procedures

NOTICE

For components included in Section 5.2 “Component Specific Torque Procedures” you must follow the requirements in both Sections 5.1 and 5.2.

NOTICE

Important!

- ▶ It is important for optimum torquing to ensure that chamfered or beveled edges are placed correctly during installation.
- ▶ Inspect washers and nuts for the correct correlation of assembly to ensure correct surface contact interface!

5.1 General Installation Requirements for ALL Torqued Bolts

1. Ensure threads are clean and free of debris or burrs, tap threads as needed. Use appropriate cleaner on all threads of blind holes to remove contaminants.
2. Check the flange surfaces for burrs and high spots or any foreign debris and correct as needed.
3. Where lubrication is required per Section 7.1 “Component Specific Torque Chart” adhere to the following:
 - Ensure 360° coverage of all threads on bolt.
 - Apply lubrication to only the stud/bolt threads unless specified in Section 7.1 or 7.2.
 - Ensure the mating surfaces between nut, washer, & bolted flange surface is clean. Failure to do so affects the holding friction.

NOTICE

The preferred lubrication brand is Jet Lube Marine Moly brand MoS2 aerosol or equivalent; and must allow uniform application and distribution on threads.

4. Verify proper orientation of washers and nuts to ensure correct surface contact.
5. Thread fasteners by hand for the first 4 threads to ensure proper thread engagement.
6. Torque bolt to value in Section 7.1 “Component Specific Torques” or if not listed there, refer to Section 7.2.1 (metric), 7.2.2 (inch coarse & fine) or 7.2.3 (electrical connections; unless otherwise indicated on the appropriate assembly drawing or specifications (which supersedes values defined in this document).
7. All fasteners must be marked with a single line with an oil-based paint marker. See Figure 1:
 - Down the stud to the nut & down the nut side, across the washer and onto the clamped part
 - Across the head downside of the bolt head, across the washer and onto the clamped part

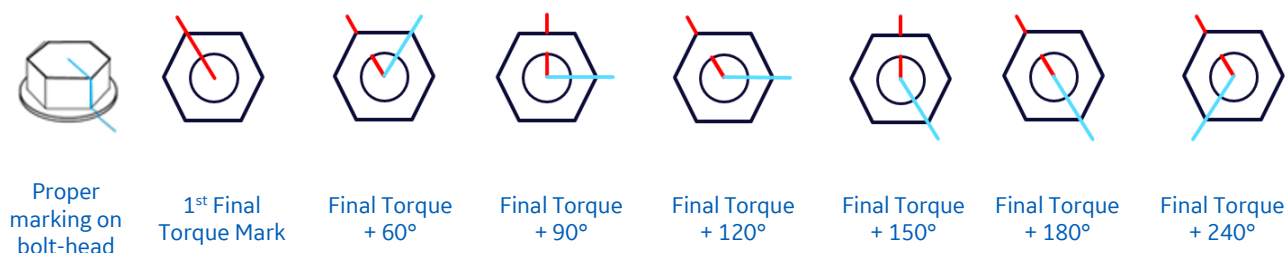


Figure 1: Marking examples for Tower Bolts, Rotor Bolts & Blade Bolts

5.2 Component Specific Torque Procedures

Note: Certain connections have torque requirements in addition to those in Section 5.1 “General Installation Requirements for ALL Torqued Bolts”. It is necessary to follow the directions in both Section 5.1 and the relevant Section below.

5.2.1 Tower Flange to Yaw Bearing Joint

NOTICE

When landing the machine head onto the tower top, slowly lower the machine head onto the tower top until the joint is flush without gaps. Only hand tighten bolts as the crane lowers the machine head onto the tower flange. DO NOT use an impact to close remaining misalignment between the machine head and tower top flange. This could degrade the life of the bolts and possibly break them off into the yaw bearing. If there is ANY damage to the hardware, a PAC case must be submitted for engineering disposition.

Critical to Quality

The machine head flange may NOT be left partially torqued. The full sequence of torque + angle must be completed when the machine head is installed.

For all turbine configurations with Torque + Angle tower to yaw bearing tightening procedure:

1. Remove water and ice from the yaw bearing/tower flanges and yaw bearing threaded holes before installation.
2. 1st Pass, Impact Installation:
 - Stab all bolts in the flange and ensure the flanges are flush.
 - Lower the machine head flange onto the bottom tower flange to make contact.
 - Hand tight all bolts around the flange.
 - Impact 5 bolts in a row with 3 groups evenly distributed over the flange (15 total), as shown in Figure 6, shown below, with the blue pattern.
 - Release 90% gross weight (full machine head weight minus rigging) from the crane.

- Sequentially impact all the bolts around the entire flange including the original 15.

NOTICE

Target torque applied in 1st Pass is ~1000Nm. Rated impact fastening torque must not exceed 2000Nm. Pre-torque must not exceed 1500Nm if using electric or hydraulic torque wrench.

3. 2nd Pass, Initial Torque Installation:

- Apply torque per Section 7.1, with approved tooling, to 5 bolts in a row with 3 groups evenly distributed over the flange (15 total), as shown in Figure 6 with the blue pattern.
- Sequentially torque all of the bolts around the entire flange including the original 15.

NOTICE

When applying sequential torque or sequential torque + angle with multiple torque devices, ensure the devices are evenly spaced around the joint. Ex: 2 torque devices 180 degrees apart, 3 torque devices 120 degrees apart.

- Apply a “1st Final Torque Mark” torque mark per Figure 1 to all bolt heads around the flange.
- Ensure the flange faces are fully seated with no gaps and without any loosening

NOTICE

A gap is defined as ANY separation between flanges. Example: 0.5mm of space is considered a gap and therefore ANY space is considered a gap.

A gap is NOT larger than a sliver of light and can only be judged when the Machine Head is fully seated on the tower top flange.

- Perform a 100% torque check on the first 6 torqued bolts. If these 6 bolts pass the torque check and if NO gaps exist of any size, move onto the next torque and angle step.
- If gaps exist of any size or the 6 bolts DO NOT pass the torque check, then repeat the torque application sequentially around the joint and repeat the 6-bolt torque check.
 - Apply a new torque mark to all bolts if this additional pass is applied.
- If gaps of any size exist or the 6-bolt torque check fails a 2nd time after this 2nd additional pass of torque submit a PAC case for engineering disposition. DO NOT move onto the next torquing step and do NOT install the hub or blades until the PAC case is closed.

4. 3rd Pass, Torque + Angle:

- Apply torque + angle per Section 7.1, with approved tooling one bolt at a time per tool in use with torque and angle on the same bolt, sequentially until all bolts have torque and angle. If a hydraulic tool is used (as may be needed behind the tool hoist), torque and angle of MUST be applied one bolt at a time, NOT just torque.

**Critical to Quality**

Blades may NOT be installed before the machine head connection has been fully torqued by the required torque + angle as described above.

For all other 1&2MW variants (torque only):

1. Remove water and ice from the yaw bearing/tower flanges and yaw bearing threaded holes before installation.
2. 1st Pass, Impact Installation:
 - Stab all bolts in the flange and ensure the flanges are flush (lower the machine head flange onto the bottom tower flange to make contact).
 - Hand tight all bolts around the flange.
 - Impact 5 bolts in a row with 3 groups evenly distributed over the flange (15 total), as shown in Figure 6, shown below, with the blue pattern.
 - Release 90% gross weight (full machine head weight minus rigging) from the crane.
 - Sequentially impact all the bolts around the entire flange including the original 15.

NOTICE

Target torque applied in 1st Pass is ~1000Nm. Rated impact fastening torque must not exceed 2000Nm. Pre-torque must not exceed 1500Nm if using electric or hydraulic torque wrench.

3. 2nd Pass, Torque Application:
 - Apply torque per Section 7.1, with approved tooling, to 5 bolts in a row with 3 groups evenly distributed over the flange (15 total), as shown in Figure 6 with the blue pattern.
 - Sequentially torque all the bolts around the entire flange including the original 15.

NOTICE

When applying sequential torque with multiple torque devices, ensure the devices are evenly spaced around the joint. Ex: 2 torque devices 180 degrees apart, 3 torque devices 120 degrees apart.

- Apply a “1st Final Torque Mark” torque mark per Figure 1. Mark to all bolt heads around the flange.
- Ensure the flange faces are fully seated with no gaps and without any loosening

NOTICE

A gap is defined as ANY separation between flanges. Example: 0.5mm of space is considered a gap and therefore ANY space is considered a gap.

A gap is NOT larger than a sliver of light and can only be judged when the Machine Head is fully seated on the tower top flange.

- Perform a 100% torque check on the first 6 torqued bolts. If these 6 bolts pass the torque check and if NO gaps exist of any size, move onto the next torque and angle step.
- If gaps exist of any size or the 6 bolts DO NOT pass the torque check, then repeat the torque application sequentially around the joint and repeat the 6-bolt torque check.
 - Apply a new torque mark to all bolts if this additional pass is applied.
- If gaps of any size exist or the 6-bolt torque check fails a 2nd time after this 2nd additional pass of torque submit a PAC case for engineering disposition. DO NOT move onto the next torquing step and do NOT install the hub or blades until the PAC case is closed.



Critical to Quality



Blades may NOT be installed before the machine head connection has been fully torqued as described above.

5.2.2 Typical Tower Flange

1. Remove water and ice from the bolts, bolt holes and the tower flange before installation.
2. Apply pre-torque of ~1000Nm to all bolts before proceeding with next steps. This step can be skipped for M36 or M39 bolts using torque only.
3. First round installation torque: Start with 3 groups evenly distributed over the flange. Each group with 5 bolts in a row (Blue pattern in Figure 6). Following bolts may be installed in sequence.
4. Second round installation step (torque or angle, as applicable) can be applied in sequence.
5. Use a backup wrench to prevent bolt rotation at any time when applying nut rotation angle.
6. Ensure comprehensive documentation of all tightening steps, tool parameters and settings for all bolts to fulfill code requirements. Recording of measured torque and angle values preferable.

NOTICE

Different tower flanges require different torque values and sockets!

- ▶ Ensure correct torque values are selected from the component torque chart in Section 7.1.
- ▶ Ensure socket geometry matches the nut geometry (Hex or 12-point/bi-hex/double-hex)
- ▶ Tower flange bolts with 12-point nut geometry might require non-standard backup wrenches due to reduced bolt spacing and different nut and bolt head A/F sizes

NOTICE

Storage conditions:

- ▶ Ensure dry storage of tower flange bolt sets until installation, particularly of lubricated parts.
- ▶ Opened boxes must be closed and foil must be re-sealed to prevent any ingress of water.
- ▶ Condition of lubrication should be checked before installation as Lubrication is susceptible to change by water and ice. Do not install in case of nameable water ingress.

NOTICE

Tower flange bolt installation:

- ▶ Bolt rotation must be prevented during application of the angle at torque+angle assembly at any time.
- ▶ For bihex/12-point bolts it is mandatory to maintain the orientation of bolt heads flat-to-flat through the torquing process.
- ▶ Double sided reaction arm (symmetrical torque arm) is recommended to ensure safe handling and operation (EHS), Mandatory for bolt sizes M56 and larger.
- ▶ During installation, minimize time of exposure of the nut to water, as possible. Remove water and ice of the bolted connection before installation.



Figure 2: Tower flange bolts, Boxes to be sealed against water ingress when opened



Figure 3: Exposed storage of lubricated nuts to be avoided (Minimize impact of water and ice)



Figure 4: Double sided reaction arm (example)

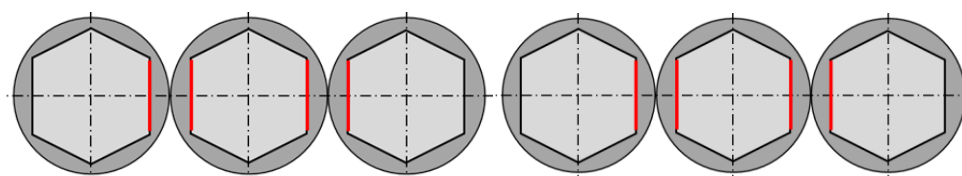


Figure 5: Orientation of bolt heads always flat-to-flat

5.2.2.1 Use of Electrical/Hydraulic Torquing Tools with Automatic Angle Measurement

When electrical/hydraulic torquing tools with automatic angle measurement are used, the torque to be applied in the 1st torquing step can be selected between the Min and the Max values according to the table below. After that, application of 100% of the torque and application of the nut rotation angle can be done in one work step for each bolt.

Without automatic angle measurement the torque applied in the 1st torquing step should be 100% of the value stated in the bolt torque specification (below table is for only tower L flanges).

	1 st torquing step		2 nd torquing step (nut rotation angle)
	Minimum (~85%)	Maximum (100%)	
M36	1600Nm	1920Nm	1920Nm + 90°
M39	2000Nm	2400Nm	2400Nm + 90°
M42	2600Nm	3100Nm	3100Nm + 90°
M48	3800Nm	4450Nm	4450Nm + 90°
M56	5800Nm	6850Nm	6850Nm + 90°
M64	8700Nm	10300Nm	10300Nm + 90°

Table 3: Torquing Steps

Torque marking to be done in single line, after torque + angle achieved when Torque+ Angle tool used.

If not, two markings to be done when the normal tool used as defined in Section 5.1.

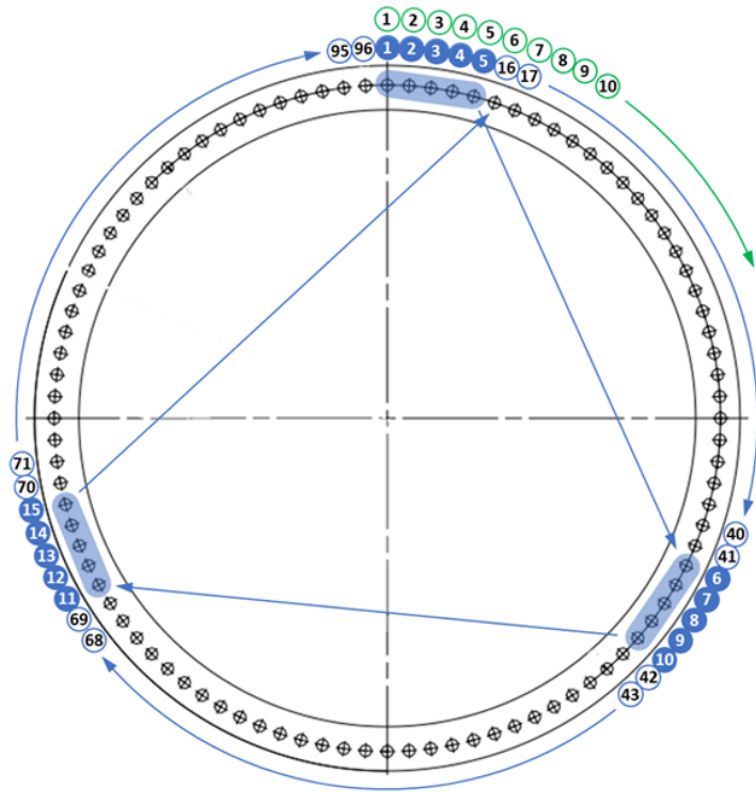


Figure 6: Example for typical tower flange numbering scheme (green) and sequence for first round installation torque (blue). Applicable for tower-to-tower L-flanges and to yaw bearing.

5.2.3 Typical Hub Adapter (Sequential Circular/Pattern)

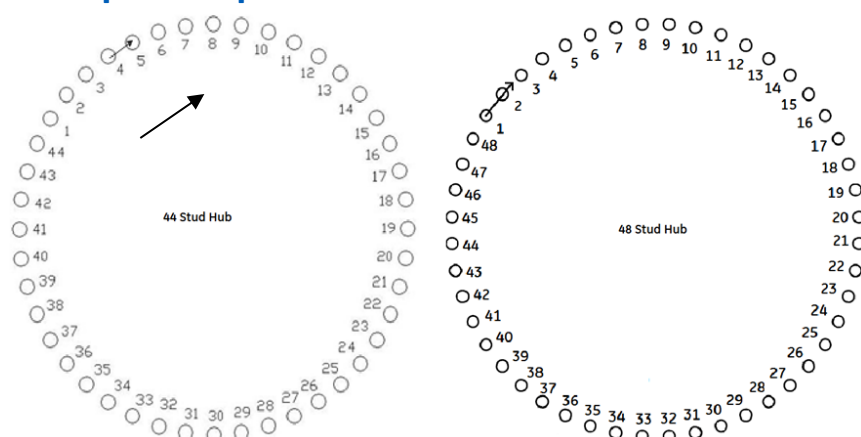


Figure 7: Typical hub adapter numbering scheme & torque sequence for 44 and 48 stud hubs.

5.2.3.1 Installation of 44 Stud Hub (1.5, 1.6-77/82.5/87/91/97, 1.68-82.5, 1.85-82.5/87)

1. Maintain a minimum of 90 % of the rotor weight on the crane.
2. Align the main shaft and rotor/hub with the crane and impact all nuts to ensure there is no gap between the flanges prior to tightening the nuts.
3. Ensure the hub bolts are properly clocked with the main shaft rotor disk.
4. Install and tighten all accessible nuts to the initial torque value as referenced in Component specific torque chart in section 7.1.
5. Release the crane.
6. Engage the high speed rotor lock when installing washers and nuts (The correct clocking will allow the Low speed Rotor lock to be engaged with the walkway frame (safety rail) exactly at Top Dead Center i.e., 12 O'clock position).
7. Slow roll the rotor/hub either CW or CCW to apply initial torque to the remaining nuts.
8. Once initial torque of all the nuts is complete, sequentially re-torque all the nuts to the initial torque value.
9. Apply a paint marking for the initial pre-torque after the re-torque is complete and write on the nacelle wall initial torque value, the date and technician's initials using the same color marker.
10. Roll the rotor/hub either CW or CCW to sequentially apply the final angle of degrees to all nuts as per Component specific torque chart in section 7.1.
11. Apply a paint marking for the final angle and write on the nacelle wall final angle value in degrees, the date and technician's initials using the same color marker. Ensure you are using a different color than that used for initial pre-torque marking.
12. Perform 10 % torque check to the maintenance torque value as referenced in Component specific torque chart in section 7.1.

5.2.3.2 Installation of 48 Stud Hub (2500 hub, 1.6-100, 1.7-100 and 1.7-103):

1. Maintain a minimum of 90 % of the rotor weight on the crane.
2. Align the main shaft and rotor/hub with the crane and impact all nuts to ensure there is no gap between the flanges prior to tightening the nuts.
3. Ensure the hub bolts are properly clocked with the main shaft rotor disk.
4. Install and tighten between 18 and 24 nuts to the initial torque value as referenced in Component specific torque chart in section 7.1.
5. Release the crane.
6. Engage the high speed rotor lock when installing washers and nuts (The correct clocking will allow the Low speed Rotor lock to be engaged with the walkway frame (safety rail) exactly at Top Dead Center.i.e., 12 O'clock position).
7. Slow roll the rotor/hub either CW or CCW to apply initial torque to the remaining nuts.
8. Once initial torque of all the nuts is complete, sequentially re-torque all the nuts to the initial torque value.
9. Apply a paint marking for the initial pre-torque after the re-torque is complete and write on the nacelle wall initial torque value, the date and technician's initials using the same color marker.
10. Roll the rotor/hub either CW or CCW to sequentially apply the final angle of degrees to all nuts as per Component specific torque chart in Section 7.1.
11. Apply a paint marking for the final angle and write on the nacelle wall final angle value in degrees, the date and technician's initials using the same color marker. Ensure you are using a different color than that used for initial pre-torque marking.
12. Perform 10 % torque check to the maintenance torque value as referenced in Component specific torque chart in section 7.1.

5.2.3.3 Installation of 48/52 Stud Hub (2570/2570RE hub, 2.0 MW to 2.8 MW ~ 107, 116, 127, 132):

1. Maintain a minimum of 90 % of the rotor weight on the crane.
2. Align the main shaft and rotor/hub with the crane and impact all nuts to ensure there is no gap between the flanges prior to tightening the nuts.
3. Ensure the hub bolts are properly clocked with the main shaft rotor disk.
4. Install and tighten at least 32 nuts to the initial torque value as referenced in Component specific torque chart in section 7.1.
5. Release the crane.
6. Engage the high speed rotor lock when installing washers and nuts (The correct clocking will allow the Low speed Rotor lock to be engaged with a Rear Hatch or the walkway frame (safety rail) exactly at Top Dead Center .i.e., 12 O'clock position).
7. Slow roll the rotor/hub either CW or CCW to apply initial torque to the remaining nuts.
8. Once initial torque of all the nuts is complete, sequentially re-torque all the nuts to the initial torque value.
9. Apply a paint marking for the initial pre-torque after the re-torque is complete and write on the nacelle wall initial torque value, the date and technician's initials using the same color marker.

10. Roll the rotor/hub either CW or CCW to sequentially apply the final angle of degrees to all nuts as per Component specific torque chart in section 7.1.
11. Apply a paint marking for the final angle and write on the nacelle wall final angle value in degrees, the date and technician's initials using the same color marker. Ensure you are using a different color than that used for initial pre-torque marking.
12. Perform 10 % torque check to the maintenance torque value as referenced in Component specific torque chart in section 7.1.
 - For 2570 Front Entry Hub (2.X 107/116) ONLY, perform 100% one-time torque check as described in section 5.2.3.3.1

5.2.3.3.1 Specific Installation Torque Check for 2570 Front Entry 2.X 107/116 Hub Adaptor

Note: The following procedure is for installation of the rotor to main shaft mounting connection and is a onetime activity to be performed during rotor installation. Refer to 7.1 Component Specific Torque Chart when performing subsequent maintenance activities.

1. Change torque setting to the value of 4500 Nm and torque check all hardware in circular pattern.
Note: Nut rotation is limited to 60 degrees during this step, torque process should be terminated if the nut rotates 60 degrees and final tool torque (or pressure) should be recorded.
2. If all connections reach a final tool torque (or equivalent pump pressure) of 4200Nm or higher and rotate less than or equal to 60 degrees, apply a new torque stripe across stud, nut, washer, and rotor lock disc face. No further action is necessary.
3. If any nut reaches 60 degrees and the tool torque is less than 4200Nm, enter a PAC or RCS case with final tool torque and degrees of rotation for all 48 connections to determine if the hardware should be replaced and retorqued.

5.2.4 Typical Rotor Blade to Pitch Bearing Stud

NOTICE

Different blades can require different torque values. Ensure correct torque values are selected from Table 5 and Table 6.

For the initial tightening sequence, refer to the 1&2MW Platform Wind Turbine Generator Installation Manual.

5.2.4.1 Tools



CAUTION



Prevent Personnel Injury!

All tools must be approved by the supplier for described process with maximum torque limit per Section 7.1.1 applied at the nut.

- **Sockets:** Thin wall sockets with a maximum OD of 83mm are needed for installation and maintenance of the blade to pitch bearing joint due to tight bolt spacing. Standard sockets with 1-1/2" square drive will have interference.
- **Tightening tools, reaction arm:** Tools with an enlarged reaction arm required during all installation and during maintenance where a torque greater than 2000Nm is required. Reaction arm shall react minimum on the 2nd bolt adjacent to the bolt being tightened (1 free bolt between). Advantage of this is a reduced tightening torque and reduced lateral force on the bolts used for the reaction point.
- **Tool calibration:** Tightening tools must be calibration according to Section 4.3.

5.2.4.2 1&2MW-100/103/107/116/127/132 only (Cross-Over Pattern) - Down-Tower Assembly

Follow pattern as illustrated in groups of three:

- 10 o'clock 3 bolts
- 2 o'clock 3 bolts
- 8 o'clock 3 bolts
- 4 o'clock 3 bolts
- 9 o'clock 3 bolts
- 3 o'clock 3 bolts

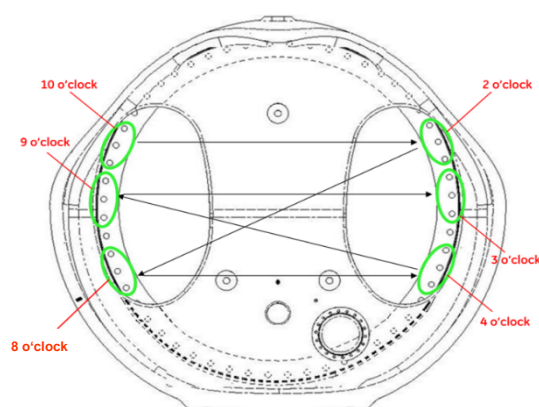


Figure 8: Typical rotor blade to pitch bearing stud torque sequence (1&2MW-100/103/107/116/127/132)

For installation and major component exchange, prior to applying pre-torque value, all nuts must be installed and impacted in the specified pattern on all three blades.

1. Apply pre-torque (check table in Section 7.1.1 under blade type) to these bolts and to remaining bolts in between the pattern. Continue pattern until all bolts are pre-torqued.
 2. Pitch the blades to expose remaining studs.
 3. Continue pattern until all bolts are pre-torqued.
 4. After all bolts have been pre-torqued, apply the additional specified nut turn from Section 7.1 of the torque chart.
 5. Continue pattern to torque the exposed bolts between the 8 to 10 and 2 to 4 o'clock positions.

5.2.4.3 1&2MW-100/103/107/116 only (Circular Pattern) - ONLY Up-tower Assembly

NOTICE

Steps 3, 4 and 5 need to be performed with the blade pointing downwards.

1. Install all exposed blade nuts using impact gun (achieve the final pre-torque value specified in the Bolt Torque Specification) – Do not pitch in this position.
2. For all installation procedures point the blade downwards before applying pre-torque and nut turn (Steps 3 and 5).
3. Install all remaining blade nuts using impact gun (achieve the final pre-torque value specified in the Bolt Torque Specification).
4. A) For flatwasher configuration: Loose nuts again (one by one) and apply the specified pre-torque to all the bolts in the circle (see Figure 9). Mark the starting point following a sequential order and continue to re-apply the pre-torque one by one to the bolts in the initial half of the circle (Figure 9). Mark all pre-torqued studs (white), to ensure that 2 circles are torqued.
5. B) For wedgelock configuration (**Loosening of nuts is not required**) : Apply the specified pre-torque to all the bolts in the circle (see Figure 9). Mark the starting point following a sequential order and continue to re-apply the pre-torque one by one to the bolts in the initial half of the circle (Figure 9). Mark all pre-torqued studs (white), to ensure that 2 circles are torqued.
6. Apply the specified nut turn to all the bolts in one circle (see Figure 10), mark all studs as usual.

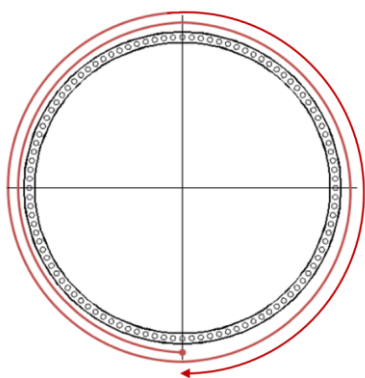


Figure 9: Step 4 - apply pre-torque to all the studs in two circles

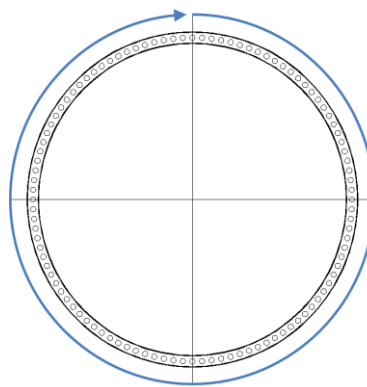


Figure 10: Apply nut turn to all the studs in one circle

5.2.4.4 1&2MW -77/82.5/87,91,97 (Star Pattern/Rolling Triangle)

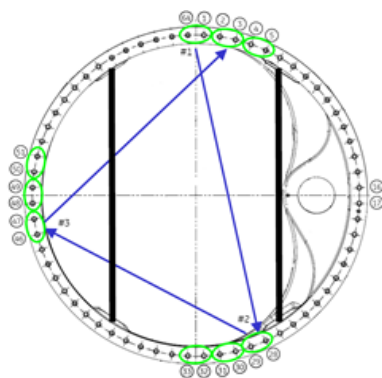


Figure 11: Typical rotor blade to pitch bearing stud torque sequence of 64 bolts (1&2MW-82.5/87/91/97)

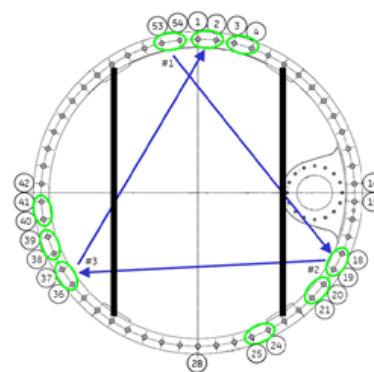


Figure 12: Typical rotor blade to pitch bearing stud torque sequence of 54 bolts (1&2MW-77)

5.2.5 Typical Pitch Bearing to Hub (Down Tower)

Note: The following procedures are for installation of the pitch bearing to hub when the pitch bearing is unloaded (i.e., rotor blade is not installed onto the pitch bearing)

1. Complete installation torque on nuts at 12 o'clock, 6 o'clock, 3 o'clock and then 9 o'clock positions, ensuring that the reaction arm is reacting off a previously torqued nut. If no previously torqued nut is available, a dummy stud must be used. The dummy stud must be replaced with a new stud after the reaction arm reacts off the dummy stud.

For installation torque refer to 7.1 Component Specific Torque Chart.

2. Immediately after torquing each nut, apply a blue paint stripe being sure to mark the nut, washer and bearing face.
3. Continue applying installation torque to remaining nuts in a circular pattern.
4. Visually inspect each nut and ensure a minimum of 1 exposed thread past the nut.

5.2.5.1 Specific Torque Check for M30 Pitch Bearing to Hub Bolts

Note: The following procedure is for installation of the pitch bearing to hub and is a onetime activity. Refer to 7.1 Component Specific Torque Chart when performing subsequent maintenance activities.

1. Change torque setting to 1700 Nm with 0 angle and torque check all hardware in circular pattern. (Note: Nut rotation is limited to 60 degrees during this step, torque process should be terminated if the nut rotates past 60 degrees).
2. If final position is 60 degrees or less past the installation torque mark after the 1700 Nm torque check, apply a new torque stripe across stud, nut, and washer onto bearing face.
3. If hardware rotation exceeds 60 degrees the hardware (stud, washer, and nut) must be replaced and re-torqued (Installation torque and check torque).

5.2.5.2 Specific Torque Check for M36 Pitch Bearing to Hub Fasteners

Note: The following procedure is for installation of the pitch bearing to hub mounting connections and is a onetime activity. Refer to 7.1 Component Specific Torque Chart when performing subsequent maintenance activities.

1. Change torque setting to the value of 3200 Nm with 0 angle and torque check all hardware in circular pattern.
2. (Note: Nut rotation is limited to 60 degrees during this step, torque process should be terminated if the nut rotates past 60 degrees).
3. If final position is 60 degrees or less past the installation torque mark after the maintenance torque check, apply a new torque stripe across stud, nut, and washer onto bearing face.
4. If hardware rotation exceeds 60 degrees, the hardware (stud, washer, and nut) must be replaced and re-torqued (Installation torque and check torque).

5.2.6 Tightening of the Tensioning Nuts for the Generator Elastic Mounts

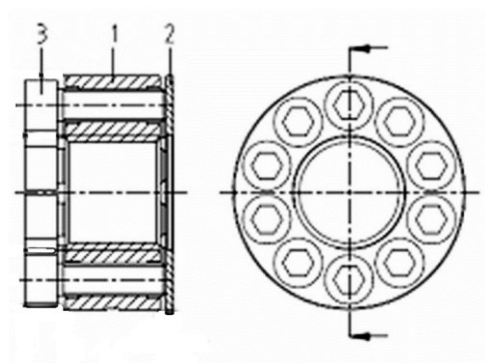


Figure 13: Assembly situation

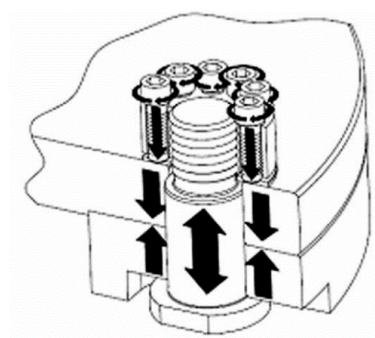


Figure 14: Functional principle of the tensioning nut

The complete tensioning nut consists of a nut (pos. 1), washer (pos. 2) and pressure bolts (pos. 3).

The pressure bolts (Figure 13, pos. 3) in the tensioning nut may not protrude from the nut on the pressure side (see Figure 15).

1. **Slip on the washer (Figure 13, pos. 2) and then screw on the nut (Figure 13, pos. 1) and fasten finger tight.**
2. **Next screw out the nut (Figure 13, pos. 1) by approx. a quarter turn.**
(This results in a gap of approx. 1 mm).

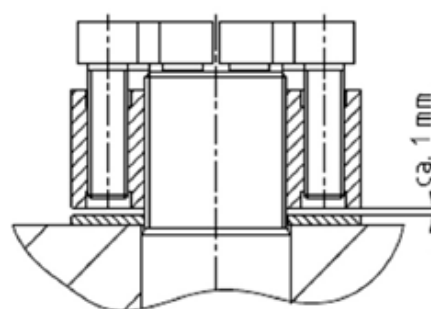


Figure 15: Screwing out the nut

NOTICE

All components must rest against each other without a gap.

3. First, fasten the M 10 pressure bolts marked in Figure 16 (pos. 1, 2, 3, 4) finger tight.
4. Then, tighten the M 10 pressure bolts to 50 % of final value.
5. In the next pass tighten the M 10 pressure bolts to the requisite torque shown in 7.1 Component Specific Torque Chart.
6. Next, tighten all eight pressure bolts one after another in several passes, until all bolts have been tightened to the requisite torque shown in 7.1 Component Specific Torque Chart
7. Apply torque marking on the individual pressure bolts and the entire assembly to the clamped flange per Section 5.1

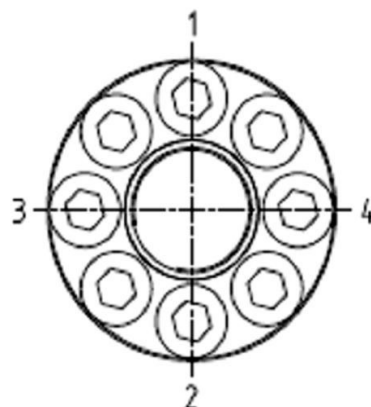


Figure 16: Tightening the pressure bolts

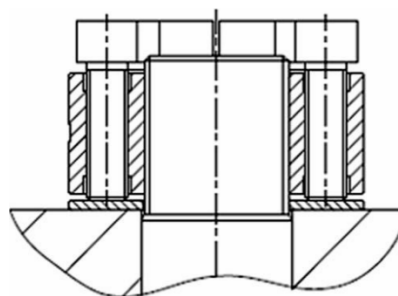


Figure 17: Completely assembled tensioning nut

NOTICE

The pressure bolts may not abut with the bearing face after the assembly (Figure 17).

6 Testing, Torque, and Inspection Instructions for Bolted Connections

6.1 Visual Inspections of Torqued/Tensioned Fasteners

Inspect each fastener for the following conditions:

- **Corrosion** – note and report to GE Representative
- **Defects** – Check for unusual or unexpected wear, fretting, flaking, polishing or particle generation (metal or otherwise) around each fastener's individual components and mating surfaces
- **Movement** – Inspect existing mark on torqued fastener for:
 - If there is no visible indication of movement, no action is required.
 - If most recent torque markings are not aligned, add witness mark to the flange aligned with previous torque mark on the nut/bolt head to record total fastener movement after torquing. See Figure 18 below as an example.
 - If there is visible movement less than 20 degrees ($< 20^\circ$) or the mark/torque stripe is illegible or missing, it is necessary to perform physical torque check on that individual fastener to maintenance values stated in the Section 7.1.
 - If there is movement greater than 20 degrees ($> 20^\circ$) when performing a visual inspection or torque check (applicable to step 2), it is necessary to torque check that fastener and the 2 adjacent fasteners on the joint. Continue torque checking adjacent fasteners until no movement is observed.
 - If at any point through visual inspection or physical torque checks more than 10% of the fasteners on the joint exhibit greater than 20 degrees ($> 20^\circ$) movement, a 100% torque check of that joint is necessary.
 - If a 100% torque check of the joint is necessary, notify a GE representative and enter a PAC case detailing the movement.
 - It may be necessary to mark the stud or bolt to indicate a 100% visual check was completed. Refer to section 6.1.1 for a list of joints that require validation of visual inspection.

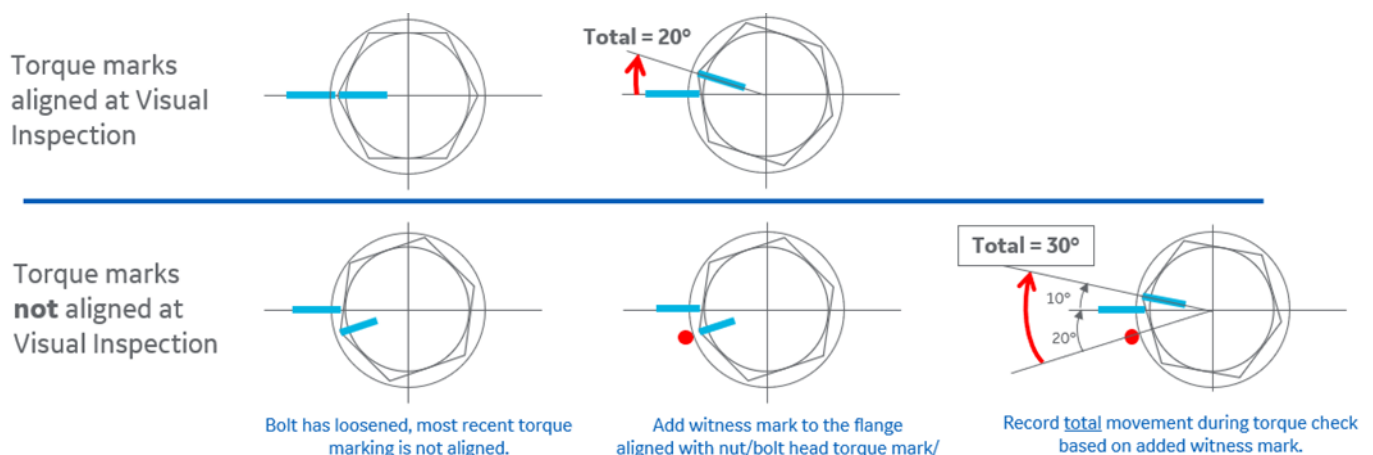


Figure 18: Torque marking – Record total fastener movement during torque check

6.1.1 Visual Inspection validation

The following joints must have a mark added to the fastener to indicate completion of the visual inspection. In cases where marking the individual fasteners is not practical, a sign off on the turbine in the location of the inspected joint is acceptable:

1. Pitch Bearing To Hub
2. Main Shaft to Hub Adapter
3. Connections between Generator Frames and/or Bedplates
4. Gearbox to Bedplate Support

Mark each fastener with a paint pen, of the color corresponding to the inspection year, to indicate visual inspection was performed per Figure 19.

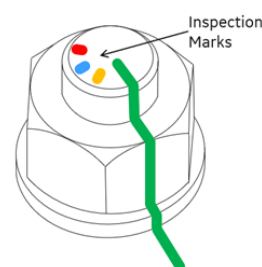


Figure 19: Example of markings used to show visual inspections

6.2 Inspection of Fasteners with Thread Locking Compound

Note: This process does not apply to any fastener larger than M24. Any critical joint covered in Section 7.1 of this document does not apply either (unless specifically identified).

1. Perform visual inspection of fastener per Section 6.1. (Note: All glued (thread locker – i.e., LOCTITE) fasteners should have torque marks applied during manufacture or commissioning.)
2. Check hand tight (or small hand wrench far below fastening torque) to ensure that bolt has not sheared in place.
3. If fastener shows no indications of movement from steps 1 and 2, no further actions are required.
4. Corrective actions if fastener has moved:
 - Remove existing fastener.
 - Re-use or replace fastener depending on criticality or condition of fastener.
 - Apply adequate thread lock compound (if not pre-applied on new fastener).
 - Reinstall fastener and apply proper torque per Section 7.1 of BTS or assembly drawings.
 - Mark newly applied fastener per Section 5.1.

6.3 How to Determine Torque Associated with Pre-Loads

Depending on the regional installation and major component exchange requirements, use this procedure to determine torque associated with pre-loads (if applicable). Field adjustable, calibrated torque wrenches must be used to tighten all tower flange bolts. Steps 1 through 7 must be performed at a minimum daily for each bolt size and grade.

1. A calibrated preload device, capable of use with the installation hardware shall be used. (i.e., Skidmode test)

2. Follow all guidelines specified by the manufacturer of the device.
3. Select at random, three-bolts, washers, and nut (bolt set-up)
4. Use a steel spacer that matches the thickness of bolted tower flange for use with the tensioning pre-load device.
5. Determine the torque value required for each of the bolt, washer, and nut sets selected to obtain the pretension value specified in the Component Specific Torque Chart.
6. Average the resulting three torque values and use this average for all tensioning for that day unless the restrictions imposed in step 8 occur.
7. Torque associated with preload results values must be recorded by installer in 3 locations: Daily Log, ICL and Tower wall next to the flange bolts for future reference with date to indicate proper torque was completed.
8. This procedure will be repeated if the temperature changes by more than 20 degrees or weather conditions change (rain stop and start, etc.).

6.4 Torque Controlled and Torque Rotation Angle Controlled Bolted Connections (excluding blade bolts) - Installation Inspection and Maintenance

6.4.1 Standard Torque Check Procedure

1. For bolted joints where a sample of physical torque checks is required, torque check a minimum of 10 % of the bolts evenly distributed over the bolt reference circle after the rotor has been installed and the complete compression load is on the tower. Rotate through bolts for each maintenance cycle until all bolts have been torque checked. For 10% installation checks, use a different tool than was used to apply initial torque + angle.
2. Without loosening the bolt apply the appropriate torque value.
3. If any fastener turns more than 20 degrees ($>20^\circ$), it is necessary to torque check that fastener and the 2 adjacent fasteners on the joint. Continue torque checking adjacent fasteners until no movement is observed. If during an installation inspection any bolt is found loose or not torque marked, all bolts in the circle must be inspected and torqued as required.
4. Mark a line on the bolt with a waterproof paint marker to indicate 10 % inspection was completed. Note for projects installation, the date, the value & name of inspector must be noted. For all level 1 components connections: Base Flange-To-Anchor bolt, Tower Flange-to-Tower Flange, Top Tower Flange-To-Yaw Bearing, Rotor-To-Hub Adaptor, Pitch Bearing-To-Hub & Rotor Blade-To-Pitch Bearing. If more than 10 % of the total count of the connection's nut/bolt exhibit movement greater than 20 degrees ($>20^\circ$), a 100% torque check of that joint is necessary. The movement angle of all fasteners that move needs to be recorded and contact your GE representative or enter a PAC case so Engineering can be informed.

6.4.1.1 Specific Replacement Criteria for 2570 Front Entry Hub (107/116) Main Shaft to Hub Hardware During Maintenance Torque Checks

A PAC case should always be entered if 10% of connections on a flange exhibit movement greater than 20 degrees, per the criteria in section 6.4.1. In addition, the hub to main shaft joint for 2570 Front Entry hub units

("2.x 107" or "2.x 116") has specific hardware replacement criteria. Please refer to the GE Operations and Maintenance Manual and Frequency Tables for the frequency of torque checks. This section applies to the hub to shaft connection, depicted below:

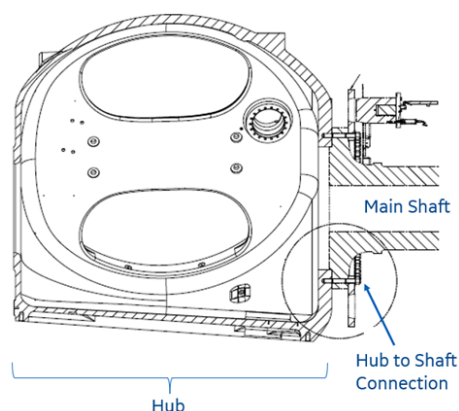


Figure 20 : Hub to shaft connection

The replacement criteria, below, does not apply to any other joint on the turbine or any other turbine platform.

Replace all 48 sets of hardware (including stud, nut, and washer) if any of the following criteria are met:

- If 5 or more nuts move greater than 20 degrees (1/3 of a flat) during torque check.
- If two or more consecutive nuts / studs are hand loose.

Replace hardware for that stud and the immediate neighbor on each side if the following criteria is met:

- Up to 4 consecutive nuts with torque mark movement greater than 20 degrees.
- One stud is hand loose.

Replace hardware (stud, washer and nut) if the following criteria is met:

- Nut(s) with torque mark(s) movement greater than 20 degrees.
- Nut(s) with torque mark(s) cumulative lifetime movement is greater than 60 degrees (example picture below)

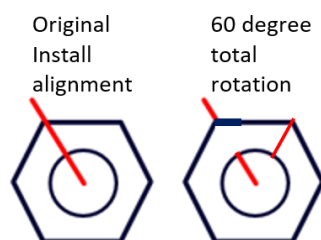


Figure 21: Torque Mark

Notes and exceptions:

1. If any of the above replacement criteria are triggered, a PAC case must be entered to notify engineering.
2. If there are ever concerns about when to replace hardware, please enter a PAC case.

3. If replacement criteria are triggered during Break in Maintenance (BIM), please enter a PAC case and a case specific engineering disposition may be issued.
4. If 5 or more connections rotate 60 degrees or more during torque check, open a PAC case, and leave turbine offline until engineering disposition.

6.4.2 Ping Test Procedure - Tower Flange Bolts

The ping test can be used in place of the standard 10 % torque check for tower flange bolts during annual maintenances every **2 years**.

NOTICE

For installation:

- The 10 % Installation checks must be completed in accordance with 6.4.1 - Torque Check Procedure for all listed connections in accordance with installation standards (Type of device, calibrations, etc...)

NOTICE

Break in Maintenance

- The 100 % Break in Maintenance (BIM) checks must be completed in accordance with 6.4.1 - Torque Check Procedure for all listed connections in accordance with BIM standards using a calibrated torque tool.

1. Starting from the 1st bolt on the flange, tap a flat on the nut with a 12-ounce ball peen hammer with enough force to hear an audible sound.
 2. Continue around the flange tapping each nut with a similar force while paying special attention for any discrepancies in the pitch of the sound.
 3. If one location has an apparent audible difference in the pitch or if movement is visually seen of the nut, tap 2-3 of the previous nuts which sounded normal and did not move to confirm a different response.
 4. Mark the suspect bolt position and continue checking the remaining bolts.
 5. If any bolts were suspected to be loose as a result of the ping test, proceed to do a 10 % torque check per Section 6.4.1 making sure to include the suspect bolts in this torque check.

6.5 Blade Bolt Connections

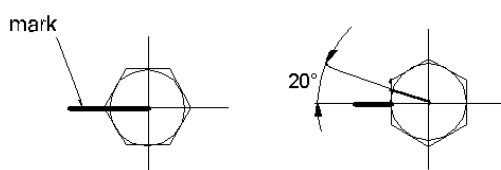


Figure 22: Installation Inspection for Blade Bolt Connection

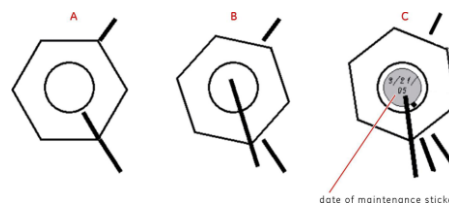


Figure 23: Example of nut markings at Installation (A), BIM (B), and maintenance (C)

6.5.1 Blade Bolt Connections – Installation Inspection

1. After completion of the rotor assembly, using a calibrated hand torque tool perform a torque inspection the bolts evenly spaced on all the three blades to ensure that no studs are not under-tensioned. Mark bolts with red paint marker.
2. If any nut moves more than 3 mm measured tangent to the washer edge (allowing for some flex of the stud shank), an error has occurred during the installation. If this occurs, then:
 - 100 % of all fasteners must be torque checked on the affected connection.
 - Any fastener that fails must be fully loosened and all installation, pre-torque and nut turn steps must be repeated on the affected nut.

6.5.2 Blade Bolt Connections - Maintenance Inspection

NOTICE

Observe the following instructions if loosening hardware:

- ▶ Ensure the mating surfaces between nut, washer and blade root are clean. Lubrication or other contaminants may reduce the holding friction.
- ▶ Each nut being torqued must be completely loosened and retightened.
- ▶ A maximum of 3 nuts per blade may be loosened at any given time.

NOTICE - Tooling

Use caution when ever perform work in the hub. When using heavy tooling ergonomic consideration should be taken into effect when attempting any work.

When using an electronic torque tool there is no need to loosen multiple fasteners. Complete loosening the nut, lubricant application (as needed), and re-torque before loosening another.

When using an electronic torque tool where the torque and angle is applied in one operation then the tool may not be removed until tensioning is completed.

NOTICE

Under normal conditions, the stud tension may decrease through seating of the stud, but it is not normal for nuts to rotate on the stud from operation.

Break in Maintenance (BIM) – Check 100 % of the blade studs, lubricating is NOT required during BIM.

Annual Maintenance - Some blade types require a torque check, i.e., application of a specific torque value, while some blade types require a re-tensioning during the 10% maintenance inspection. Please refer to Section 7.1 "Component Specific Torque Chart". 'Maintenance' column for details.

During 10% maintenance torque checks, nuts may rotate up to 30 degrees beyond previous torque mark. For blades installed with flat washers, if any hardware moves more than 30 degrees, a 100% re-tensioning (loosen and apply installation torque and angle) is required per steps 6.5.2. For blades installed with wedgelock washers, if any hardware moves more than 30 degrees, a 100% torque check at the maintenance torque value is required. If 100% checks are required, the angle of rotation for all fasteners that move must to be recorded and contact your GE representative or enter a PAC case so Engineering can be informed.

1. Visually inspect all the hardware. If any nuts have paint marks misaligned with stud (indicating rotation of nut) then document the position and estimated degree of movement of effected studs.
2. For annual maintenance, rotate through bolts for each maintenance cycle until all bolts have been torque checked equally spaced around the flange.
3. If a torque check is required, i.e., application of a specific torque value without the need to loosen hardware, for the blade type being maintained, skip to step 11.
4. If re-tensioning is required, i.e., application of a specific torque + angle after loosening and re-lubricating hardware, continue with steps 5 through 10.
5. Loosen one to three nuts around the stud circle, do NOT loosen adjacent nuts.
6. Verify stud has not backed out, if so re-install in accordance with the blade drawing.
7. Lubricate threads unless performing the Break in Maintenance, then lubrication is not necessary. If blade bolts are found without lubricate at BIM this needs to be recorded and contact your GE representative so Engineering/Product Service Engineering can be informed.
8. Hand tight the nut(s).
9. Clean mating surfaces (nut, washer, and blade root). Failure to do so affects the holding friction.
10. Apply the pre-torque and specified angle from Section 7.1 "Component Specific Torque Chart" to an accuracy of ± 5 degrees at the nut corner. For electronic torque tools apply pre-torque and angle in one operation. For tools which apply torque and angle separately, apply the pre-torque specified to all loosened nuts then apply specified angle of additional nut turn.
11. From step 3, apply the torque value specified in Section 7.1 "Component Specific Torque Chart". If the nut moves more than 30° , or half a flat, stop applying torque and see notes above.
12. Immediately after finishing torquing each nut, place the paint alignment marks across the stud end, nut, and surrounding bearing surface before proceeding to the next nut. This is necessary to avoid losing track of which nuts have been torqued as the blade is pitched for access to all studs.

13. Report the location on blade and the estimated degree of movement of any nuts that moved more than 30 degrees (or half a flat) past previous position.
14. Torque all accessible nuts, then pitch blades to gain access to the studs behind the axis boxes and/or hub webbing.
15. Repeat the steps above until the maintenance is completed.

6.6 Foundation Anchor Bolts - Axial Hydraulic Pre-tensioning (only if foundations are within GE scope)

- Pre-tension load values and inspection requirements are solely dependent on the foundation design criteria and shall not be deviated from.
 - Note: Check GE designed foundations at intervals specified in the design documents but at least annually for the first 5 years and every 3 years thereafter.
- GE designed foundations without specific pre-tensioning and/or maintenance requirements provided on the design documents will follow the procedures in 6.6.1 and 6.6.2.

NOTICE

For Non-GE foundation designs follow the manufacturer's initial tensioning and maintenance recommendations for both frequency and method.

- All tensioning and pre-tension checks are carried out in a clockwise direction around the circumference. Bolt No. 1 is located at the bottom longitudinal weld and will be identified as such during all subsequent measurements. This procedure applies for the inner and outer anchor bolts.

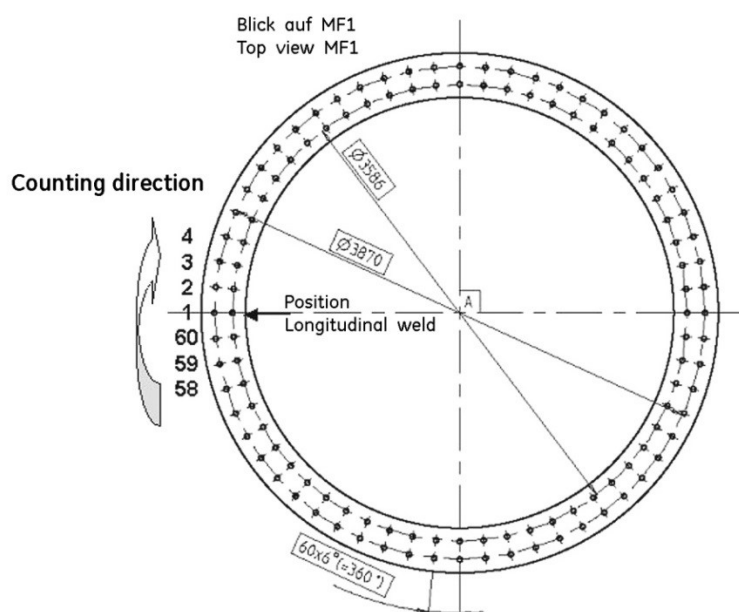


Figure 24: Numbering of anchor bolts

Document the following on the “Anchor Bolt Tension Record” for any tensioning or checks:

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- | | |
|-------------------------------------|------------------------------------|
| • Site Name | • WTG number |
| • Tool Type / Model / Serial Number | • Bolt designation |
| • Tool Calibration Date | • The final pre-tightening force |
| • Date Work Performed | • Operator |
| • Design tension (KIPS) | • Maintenance Cycle / Installation |
| • Target tension (KIPS) | • Wind speed |
| • Tool PSI | • Temperature |

6.6.1 Hydraulic Pre-tensioning of Anchor Bolts during Assembly or Refurbishment (Initial Tightening or Replacement)

- Begin at bolt # 1 and move clockwise tensioning all bolts to foundation design's installation value.
- Mark all anchor bolts immediately after tensioning.

6.6.2 Performing Anchor Bolt Maintenance Hydraulic Checks on GE Designed Foundations

- Pre-tensioning is to be performed at least annually for the first 5 years, and every 3 years thereafter or more frequently if called for in foundation design documents.
- Refer to the latest Maintenance Activity Report to determine scope of the maintenance checks (number and position of the anchor bolts).
- Maintenance pre-tension load values and inspection requirements are based on 90 % of the site-specific installation value.

NOTICE

These instructions provide an example of testing and reporting instructions. Depending on contractual and local/national requirements, this inspection may be different. Refer to your maintenance contract to determine what elements are included in the scope of supply.

- Check the pre-tension value of 10 % of the anchor bolts on the inner and outer flange and mark (at 90 % tension value). Do not release the pre-tension of the bolts beforehand!
- If any anchor bolt has any indication of movement, then perform the following:
 - a. Tension that anchor bolt to 100 % of the site-specific installation value.
 - b. Check all the anchor bolts on that flange (inner or outer) at 90 % of installation tension value.
- If 10 % of the anchor bolts on a flange (inner or outer) do not pass, then 100 % of the anchor bolts on both flanges (inner and outer) must be checked (at 90 % tension value).



Anchor Bolt Tension Record

Site Name:	WTG:
Tool type/model:	
Serial number:	
Calibration Date:	

Maintenance cycle:
Design tension (KIPS):
Target tension:
KIPS: Tool PSI:

Inside Flange			
Date Performed:			
Wind Speed:		Temperature:	
Bolt #	Movement	Bolt #	Movement
1		36	
2		37	
3		38	
4		39	
5		40	
6		41	
7		42	
8		43	
9		44	
10		45	
11		46	
12		47	
13		48	
14		49	
15		50	
16		51	
17		52	
18		53	
19		54	
20		55	
21		56	
22		57	
23		58	
24		59	
25		60	
26		61	
27		62	
28		63	
29		64	
30		65	
31		66	
32		67	
33		68	
34		69	
35		70	

Outside Flange			
Date Performed:			
Wind Speed:		Temperature:	
Bolt #	Movement	Bolt #	Movement
1		36	
2		37	
3		38	
4		39	
5		40	
6		41	
7		42	
8		43	
9		44	
10		45	
11		46	
12		47	
13		48	
14		49	
15		50	
16		51	
17		52	
18		53	
19		54	
20		55	
21		56	
22		57	
23		58	
24		59	
25		60	
26		61	
27		62	
28		63	
29		64	
30		65	
31		66	
32		67	
33		68	
34		69	
35		70	

Technician Signature:

Date:

7 Required Torque / Tension Values

7.1 Component Specific Torque Chart

NOTICE

- For any deviation from below published Torque Chart values, open a PAC case for engineer disposition.

Break-In-Maintenance NOTE

For all fasteners for any connection that becomes loose or replaced due to maintenance, repair & component upgrade, one must follow all Break-In-Maintenance requirements listed in the Bolt Torque Specifications. Any connection effected by such work should begin a new set of BIM intervals and be recorded In the WTG log.

Connection		Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
Base flange with anchor bolts		All values are site specific and provided by customer's foundation engineer						Yes
DTA extension posts		M24	32	736 Nm	Visual			Yes
PPM beams		M20	32	450 Nm	Visual			Yes
Tower flanges Tightening method: Torque	Note: Either the flange bolts or nuts MUST be pre-lubricated. Contact GE if they are not.	M36	60	2800 Nm	2800 Nm	Install per Section 5.2.2. 1st round is 35 % - 55 % of final torque. For M39 bolts only, 1st round torque is 28% - 55% of final torque.	3000 Nm	No
		M39	65	3500 Nm	3500 Nm		4000 Nm	No
		M42	70	4500 Nm	4500 Nm		5000 Nm	No
		M45	75	5500 Nm	5500 Nm	For torque only, typical bolt part numbers are 104W5057, 445W7652, or 446W4674. If detailed method is not identified within tower drawing torque only method is required.	6000 Nm	No
		M48	80	6500 Nm	6500 Nm		7000 Nm	No
Tower flanges Tightening method: Combined method (Torque + angle)	Note: Either the flange bolts or nuts MUST be pre-lubricated. Contact GE if they are not.	M36	60	1st round 1920 Nm 2nd round 1920 Nm+90° (± 10°) Max 5600 Nm See footnote [3]	BIM: 2800 Nm+ 15° (+5°) See footnote [1a] BRM: 2800 Nm or ping test	Site team to refer to site specific tower drawings for detailed information regarding torque only or torque + angle method. For torque + angle method, typical bolt part numbers are 444W4984 or 445W7294. If detailed method is not identified within tower drawing torque only method is required.	6000 Nm	No
		M39	65	1st round 2400 Nm 2nd round 2400+90° (± 10°) Max 7000 Nm See footnote [1]	BIM: 3500 + 15° (+5°) See footnote [1a] BRM: 3500 Nm or ping test		8000 Nm	No
		M42	70	1st round 3100 Nm	BIM: 4500 Nm + 15° (+5°)		10000 Nm	No

² Please refer section 5.2.2.1 for detailed bolt torque procedure

³ In case the target angle of 90° is not reached within max torque value; contact GE representative for Engineering/Product Service resolution.

^{1a} Target angle of 15° might not be reached within max torque value specified for installation. Acceptable angles between 0° and 15° (+5°)

Connection		Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
				2nd round 3100 Nm+90° (± 10°) Max 9000 Nm See footnote [3]	See footnote [1a] BRM: 4500 Nm or ping test			
		M48	80	1st round 4450 Nm 2nd round 4450Nm +90°(±10°) Max 13000 Nm See footnote [1]	BIM: 6500 Nm + 15° (+5°) See footnote [1a] BRM: 6500 Nm or ping test		14000 Nm	No
		M56	90	1st round 6850 Nm 2nd round 6850 Nm +90°(± 10°) Max 20000 Nm See footnote [1]	BIM: 10000Nm+ 15° (+5°) See footnote [1a] BRM: 10000 Nm or ping test		21000 Nm	No
		M64	100	1st round 10300 Nm 2nd round 10300 Nm +90°(± 10°) Max 30000 Nm See footnote [1]	BIM:15000Nm+ 15° (+5°) See footnote [1a] BRM: 15000 Nm or ping test		31000 Nm	No
Tower flange / Yaw bearing	1.9-127 Repower, 2.0 to 2.8 -127/132 using STUD 454W8939P001	M36	55	1500 Nm + 150°	2800 Nm	See Section 5.2.1	4500 Nm	Yes, single re-use
	2.0 to 2.8 - 107/116/127/132 BOLT	M36	55	1500 Nm + 120°	If 1500 Nm + 60°: 2500 Nm If 1500 Nm + 120°: 2800 Nm	See Section 5.2.1	4500 Nm	No
	1&2MW-100/103 BOLT	M36	55/60	1500 Nm + 120°	2800 Nm	See Section 5.2.1	4500 Nm	No

¹ In case the target angle of 90° is not reached within max torque value; contact GE representative for Engineering/Product Service resolution.

^{1a} Target angle of 15° might not be reached within max torque value specified for installation. Acceptable angles between 0° and 15° (+5°)

Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
	All other 1&2MW variants BOLT		M30	50	1850 Nm - Lubricate threads for galvanized	1850 Nm		2500 Nm	No
Yaw bearing / bedplate	2.0 to 2.8 - 107/116/127/132 with Wedge Lock washers		M36	55	1500 Nm + 120°	3800 Nm	Stud Part #: 115W2238P015 Verify the part number prior to installation or replacement	6000 Nm	No
	2.0 to 2.8 - 107/116/127/132 with Wedge Lock washers		M36	55	2400 Nm + 120°	3800 Nm	Stud Part #: 444W4065P004/P005 Verify the part number prior to installation or replacement	6000 Nm	No
	All other 1.x variants with Standard washers		M36	60/55	500 Nm + 120°	2800 Nm		4500 Nm	No
Yaw Friction System - Adjusting Screw	Puck Low Friction Coefficient		M39	60	100 Nm + 300°	100 Nm + 300°	Yaw puck piston	1000 Nm	Yes
	Puck High Friction Coefficient		M39	60	150 Nm + 180°	150 Nm + 180°	4-4-3 Spring Stack Retrofit	1000 Nm	Yes
					150 Nm + 360°	150 Nm + 360°			
Yaw puck cover plate			M36	60	2100 Nm	2100 Nm		2500 Nm	Yes
			M39	2- ³ / ₈ /60	2100 Nm	2100 Nm		2500 Nm	Yes
Yaw drive	1&2MW-100/103/107/116/127/132		M24	19	Delta MKS (black coating): 850Nm M11 (silver coating): 1000 Nm	Delta MKS, black: 850Nm M11, silver: 1000 Nm	444W5078P001: Delta MKS, Black L15DP33120M11: M11, Silver		Yes
	All other 1&2MW variants	10.9 grade	M20	17	580 Nm	580 Nm			Yes
		12.9 grade	M20	17	690 Nm	690 Nm			Yes
Generator Frame	2.0/2.3/2.8 - 107/116/127/132	to bedplate	M36	55	1500 Nm + 90°	2500 Nm		4000 Nm	No
		to frame legs	M39	60	2000 Nm + 60°	3700 Nm		4000 Nm	No
		lifting eyes	M24	36	1000 Nm	N/A	Removed after installation completed.		No
	All other 1&2MW variants		M36	60	500 Nm + 120°	2800 Nm		4000 Nm	No
Shrink disc	1.x-116 & 1.9-100 RePower		M33	50	2210 Nm	2210 Nm	Installation process to obtain flush face		Yes
	All other 1&2MW variants		M30	46	1640 Nm	1640 Nm			Yes
Main gearbox support	2.0 to 2.8 - 107/116 (i.e. split main bearing housing,		M39	60	2000 Nm + 210°	3700 Nm	Stud Part #: 115W2238P016: Shank Ø Not Tapered	6000 Nm	See “Re-use Allowed NOTE”

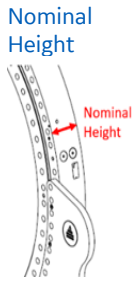
Connection		Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
	front entry hub, cap bedplate)					Verify the part number prior to installation or replacement		
	2.0 to 2.8 - 116/127/132 (i.e. rear entry hub, compact bedplate)	M39	60	2400 Nm + 240°	3700 Nm	Stud Part #: 444W7227P001: Shank Ø Is Tapered Verify part number prior to installation or replacement	6000 Nm	See "Re-use Allowed NOTE"
	1.x-116 & 1.9-100 RePower	M36	60/55	1800 Nm + 210°	2800 Nm		6000 Nm	See "Re-use Allowed NOTE"
	Machined Mounts (Square Top)	M36	55	1500 Nm + 180°	2800 Nm		4000 Nm	See "Re-use Allowed NOTE"
		M36	60	1500 Nm + 180°	2800 Nm		4000 Nm	See "Re-use Allowed NOTE"
	Cast mounts and cast w/ shim mounts (Round Top)	M36	55/60	1500 Nm + 150°	2800 Nm		4000 Nm	See "Re-use Allowed NOTE"
Main bearing pillow block	2.0 to 2.8 - 107/116 (i.e. Split main bearing housing, front entry hub, cap bedplate)	M48	75	3700 Nm + 90°	6000 Nm	Stud Part #: 103W3216P003: Shank Ø Not Tapered Verify the part number prior to installation or replacement	10000 Nm	No
				3700 Nm + 120°		Stud Part #: 444W9106P002: Shank Ø is Tapered Verify the part number prior to installation or replacement		
	2.0 to 2.8 -116/127/132 (i.e. rear entry hub, compact bedplate)	M48	75	2500 Nm + 120°	6000 Nm	Stud Part #: 444W9106P001: Shank Ø Is Tapered Verify the part number prior to installation or replacement	10000 Nm	No
	1&2MW - 100/103	M48	75	5300 Nm + 60°	6500 Nm		10000 Nm	No
	1.x-82.5/87/91/97	M39	60	2000 Nm + 150°	3500 Nm		6000 Nm	No
	1.x-116 & 1.9-100 RePower	M48	75	2500 Nm + 120°	6000 Nm		10000 Nm	No
	All other 1&2MW variants	M39	60	2800 Nm + 60°	3500 Nm		6000 Nm	No
Main bearing cover plate	2.x - 107/116/127/132	M24	36	1000 Nm	1000 Nm	Bolt part #: L114BP33080M11 Verify the part number prior to installation or replacement		Yes
	2.x - 107/116/127/132	M24	36	529Nm	529Nm	Bolt part #: 105W1931P071 Verify the part number prior to installation or replacement		Yes

Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
Main bearing locknut	SKF		M16	14	505 Nm	505 Nm	Split locknut design - (1) clamping screw in locknut		Yes
	FAG		M16	14	230 Nm	230 Nm	Slotted thread design - (4) screws inside face		Yes
	FAG		M20	17	464 Nm	464 Nm	Slotted thread design - (4) screws inside face		Yes
Nacelle Material Handling Rails	Bracket to Material Handling Rail		M12	18	70 Nm	70 Nm			Yes
Brake Caliper to Gearbox	Svendborg BSFH		M20	30	350 Nm	350 Nm			Yes
	BSFI/BSAK/Hydrep		M24	36	690 Nm	690 Nm			Yes
	1&2MW – 77, 82.5, 87, 91, 97, 100, 103		M24	36	720 Nm	720 Nm		1000 Nm	Yes
	2.x – 107/116/127/132		M36	55	2800 Nm	2800 Nm		4000 Nm	Yes
High Speed Shaft Coupling	Multiple platforms		Multiple manufactures				See supplier document and relevant GE Vernova procedure	-	Yes
Rotor lock high-speed shaft	All other 1&2MW variants		M24	36	690 Nm	690 Nm			Yes
Generator to damper mount	Normal nut		M30	46	800 Nm	800 Nm			Yes
	Tensioning nut		M10	16	26 Nm	Visual			Yes
Generator vibration damper to bedplate			M20	30	420 Nm	Visual	Use threadlock		No
Generator j-box bus bar			Use values on nameplate inside junction box. If none is provided use values listed in Section 7.2.3						Yes
Nacelle hoist emergency escape ring					60 Nm	25 Nm	Use threadlock		No
Rotor Hub to main shaft flange	1 MW – 77, 82.5, 87, 91, 97	Wedge lock	M36	55	2400 Nm + 90°	3800 Nm	1st pass: 2400 Nm 2nd pass 2400 Nm 3rd pass: 90° Stud Part #: 115W2238P003: Shank Ø Not Tapered Verify the part number prior to installation or replacement Refer to section 5.2.3.1 for detailed installation procedure	6000 Nm	See “Re-use Allowed NOTE”
	1 MW – 100,103	Wedge lock	M36	55	2400 Nm + 120°	4200 Nm	1st pass: 2400 Nm 2nd pass 2400 Nm 3rd pass: 120°	6000 Nm	See “Re-use Allowed NOTE”

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Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
	2 MW FEH-100, 107, 116						Stud Part #: 115W2238P010: Shank Ø Not Tapered Verify the part number prior to installation or replacement Refer to section 5.2.3.2, 5.2.3.3 and 5.2.3.3.1 for detailed installation procedure *Maintenance value updated from 3800Nm to 4200Nm in R21. Prior to moving to 4200Nm, ONE torque check must be done at 4500Nm per procedure in 5.2.3.3.1. All subsequent checks will be at 4200Nm		
	2 MW FEH-100, 107, 116	Wedge lock	M36	55	2400 Nm + 120°	4200 Nm	1st pass: 2400 Nm 2nd pass 2400 Nm 3rd pass: 120° Stud Part #: 444W4065P001: Shank Ø Is Tapered Verify the part number prior to installation or replacement Refer to section 5.2.3.3 and 5.2.3.3.1 for detailed installation procedure *Maintenance value updated from 3800Nm to 4200Nm in R21. Prior to moving to 4200Nm, ONE torque check must be done at 4500Nm per procedure in 5.2.3.3.1. All subsequent checks will be at 4200Nm	6000 Nm	See "Re-use Allowed NOTE"
	1MW – 77, 82.5, 87, 91, 100, 103	Standard	M36	55	1500 Nm + 90°	3000 Nm	1st pass: 1500 Nm 2nd pass 1500 Nm 3rd pass: 90° Refer to section 5.2.3.1 for detailed installation procedure	4500 Nm	See "Re-use Allowed NOTE"
	2MW REH – 116, 127, 132	Wedge lock	M39	60	2800 Nm + 180°	5000 Nm	1st pass: 2800 Nm 2nd pass 2800 Nm 3rd pass: 180°	8000Nm	See "Re-use"

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Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
							Refer to section 5.2.3.3 for detailed installation procedure		Allowed NOTE
	All other 1.x variants	Standard	M36	55	1500 Nm + 90°	3000 Nm	1st pass: 1500 Nm 2nd pass 1500 Nm 3rd pass: 90° Refer to section 5.2.3.1 for detailed installation procedure	4500 Nm	See "Re-use Allowed NOTE"
Pitch bearing to Hub		132 mm	M30	50	500 Nm + 100°	1300 Nm			No
		163 mm	M30	50	500 Nm + 140°	1300 Nm	See 5.2.5.1 for detailed installation procedure		No
		163 mm Wedge Lock	M30	50	700 Nm + 140°	1300 Nm	See 5.2.5.1 for detailed installation procedure	3000 Nm	No
		173.5 mm Wedge Lock	M30	50	700 Nm + 140°	1300 Nm	See 5.2.5.1 for detailed installation procedure	3000 Nm	No
		172 mm	M36	55	1500 Nm + 120°	2500 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check		See "Re-use Allowed NOTE"
		231 mm	M36	55	1500 Nm + 150°	2500 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Stud Length 350 mm PN - 115W2238P011 Verify the part number prior to installation or replacement Used on front entry hubs, NOT for stabilization plate or lifting lug. The pitch bearing stud part numbers ending in '2238' have a straight shank (36mm shank, same as outer diameter of threads) and do NOT have a protrusion at the end of the stud.		No
		231 mm	M36	55	1500 Nm + 150°	2500 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Stud Length 400 mm PN - 115W2238P014		No

Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
							Verify the part number prior to installation or replacement Used on front entry hubs for stabilization plate or lifting lug. The pitch bearing stud part numbers ending in '2238' have a straight shank (36mm shank, same as outer diameter of threads) and do NOT have a protrusion at the end of the stud.		
		231 mm	M36	55/60	1500 Nm + 180°	2500 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Stud Length 350 mm PN - 444W4065P002 Verify the part number prior to installation or replacement Used on front entry hubs, NOT for stabilization plate or lifting lug. The pitch bearing stud part numbers ending in '4065' have a tapered shank (smaller than outer diameter of threads) and a protrusion at the end of the stud.		See Re-use Allowed NOTE
		231 mm	M36	55/60	1500 Nm + 180°	2500 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Stud Length 400 mm PN - 444W4065P003 Verify the part number prior to installation or replacement Used on front entry hubs for stabilization plate or lifting lug. The pitch bearing stud part numbers ending in '4065' have a tapered shank (smaller than outer diameter of threads) and a protrusion at the end of the stud.		See Re-use Allowed NOTE
		231 mm	M36	60	2500 Nm + 180	2800 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Square headed bolt Length 423 mm PN - 446W2076P001		No

Connection			Size	Socket or Hex	Installation	Maintenance	Notes ²	Minimum Tool Capacity	Hardware Re-use Allowed
							Verify part number prior to installation or replacement Used on rear entry hubs only. NOT for stabilization plate location. This bolt part number has wrench flats at end of bolt.		
		231 mm	M36	60	2500 Nm + 180°	2800 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Square headed bolt Length 423 mm PN - 444W3890P001 Verify part number prior to installation or replacement Used on rear entry hubs only. This bolt part number does NOT have wrench flats at the end of the bolt.		No
	Nominal Height	231 mm	M36	60	2500 Nm + 180°	2800 Nm	See 5.2.5.2 for detailed installation procedure including 100% 3200Nm check. Square headed bolt length 473 mm PN – 447W8947P001 Verify the part number prior to installation or replacement Used on rear entry hubs only at the stabilization plate location. This bolt part number has wrench flats at end of bolt.		No
Pitch drive	8.8 grade bolt		M16	14	210 Nm	210 Nm			Yes
	10.9 & 12.9 grade bolts		M16	14	295 Nm	295 Nm			Yes
Nosecone frame to hub			M20		410 Nm	Visual			No

Table 4: Component Specific Torque Chart

Re-use Allowed NOTICE

A single re-use of these fasteners is approved if all the following conditions are met.
If any condition is not met the hardware must be replaced.

- Joints must not be loose and no more than 10 % of fasteners fail maintenance torque check.
- No indication of prior re-use (punch mark or paint). No cracks, deformation (bending or stretching), and no galling/deformation/damage on threads/washer.
- Nut must be able to be run by hand over the threads above/through/below the engaged area.
- If wedge lock washer is present, ensure washer to be installed has small serrations pointing towards the flange and nut with no cracks or deformation.
- If your turbine operates with a stiffener plate on the pitch bearing to hub joint, the studs used to attach the plate to the Bearing may not be re-used.

Required documentation: Entry made in turbine log and written message on joint indicating re-use; if studs replaced at later time, then line out. Stud ends are colored in with paint pen and if studs are exposed to weather the end must additionally be marked by hitting it with a punch to indicate re-use.

Hardware Re-use Terms Definitions:

- ▶ **Yes:** The bolt can be re-used. If bolt is corroded/damaged it must be exchanged.
- ▶ **No:** The bolt can't be reused It MUST be exchange for new hardware.
- ▶ **See “Re-use Allowed NOTE”:** If any condition is not met the hardware must be replaced.

7.1.1 Torque Chart – Rotor Blade to Pitch Bearing

Rotor Blade to Pitch Bearings with Flat Washer Hardware						
Blade Type	Fastener Size	Socket Width	Installation & BIM	Maintenance	Hardware Re-use Allowed	Notes
LM - 34.0P, 34.0P3, 37.3P, 37.3P2, 40.3P	M30	50	350 Nm + 120°	900 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	- If bolts are broken or loose to touch, re-tension entire joint. - Blades requiring a torque check, i.e., application of a single torque value, during annual maintenance do NOT require loosening and lubrication. - Note some blade types require re-tensioning at annual maintenance, denoted with ONLY a torque + angle value in the Maintenance column. - If re-tensioning is required, use Installation torque + angle values and loosening and lubrication is required. - Use MOS2 lubricant when required but ONLY on threads. - Blade torques have tolerance of ±5° - Re-use of blade studs is allowed. See "Re-use Allowed NOTE". Installation Only – 10 % Inspections - Use 1000 Nm for 1&2MW-100/103/107/116/127/132. - Use 750 Nm for All other 1&2MW variants. 10 % of the total number on a connection must be checked (minimum).
GE - 34A, 34C, 37A, 37B, 37C, 37C2, 40					Refer "Re-use Allowed NOTE"	
EW - 34A, 34C, 37A, 37B					Refer "Re-use Allowed NOTE"	
APX 70; MFG 34A; Tecsis 34A					Refer "Re-use Allowed NOTE"	
LM - 42.1P2, 44.1	M30	50	500 Nm + 150°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	
GE - 48.7C, 48.7G, 50.2G, 52.2	M36	60	500 Nm + 120°		Refer "Re-use Allowed NOTE"	
LM 48.7, 52.2P	M36	60	550 Nm + 120°		Refer "Re-use Allowed NOTE"	
LM 56.9 without 100mm Stiffener Ring	M36	60	600 NM + 120°		Refer "Re-use Allowed NOTE"	
LM 56.9 w/100mm Stiffener Ring	M36	60	600 NM + 120°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	
GE 56.9 without 100mm Stiffener Ring	M36	60	600 Nm + 120°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	
GE 56.9 w/100mm Stiffener Ring	M36	60	600 Nm + 150°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	
LM 62.2 w/ Flat Washers	M36	60	750 Nm+180°	2500 Nm	Refer "Re-use Allowed NOTE"	
GE 64.5	M36	60	600 Nm + 150°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	
LM 64.6P	M36	60	600 Nm + 150°	1600 Nm 30° rotation limit	Refer "Re-use Allowed NOTE"	

Table 5: Rotor Blade to Pitch Bearing with Flat Washers Hardware Torque Chart

Rotor Blade to Pitch Bearing with Wedgelock Washer Hardware

Blade Type	Fastener Size	Socket Width	Installation	Maintenance	Hardware Re-use Allowed	Notes
LM44.1 w/ Wedgelock Washers ^[1]	M30	50	700 Nm + 180°	1200 Nm 30° turn limit	Refer "Re-use Allowed NOTE"	NOTICE: MOS2 Lubricant is not needed for Installation or Maintenance purpose. For annual maintenance, - Do NOT loosen connections with Wedgelock washers, only perform a torque check. - Blade torques have tolerance of $\pm 5^\circ$ - Wedgelock Washers for M30: P/N 444W1057P006 - Wedgelock Washers for M36: P/N 444W1058P008
LM47.3 w/ Wedgelock Washers ⁴	M30	50	700 Nm + 270°	1600 Nm 30° turn limit	NOT Allowed	
LM48.7 w/ Wedgelock Washers ⁴	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
LM56.9 w/ Wedgelock Washers ⁴	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
GE56.9 w/ Wedgelock Washers ⁴	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
GE 62.2 w/ Wedgelock Washers	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
LM 62.2 w/ Wedgelock Washers	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
GE 64.5 w/ Wedgelock Washers	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	
LM 64.6 w/ Wedgelock Washers	M36	60	1500Nm+300°	3000Nm	Refer "Re-use Allowed NOTE"	

Table 6: Rotor Blade to Pitch Bearing with wedgelock washers Torque Chart

^[1] These configurations are mostly for repower projects. Ensure washer type provided with the hub/blade is installed and torqued per Table 6

7.2 Small Diameter Bolt Torque Values

The torque values in this Section are superseded by those specified in Section 7.1 “Component Specific Torque Chart” for any specific component and/or on the face of manufacturer component drawings if applicable.

7.2.1 Metric - Small Diameter Bolt Torque Values

Thread Size	Tightening torque							
	Bolt Property Grade							
	5.8		8.8		10.9		12.9	
	lb.-in.	N-cm	lb.-in.	N-cm	lb.-in.	N-cm	lb.-in.	N-cm
M1.6	0.8	9	1.2	13	1.8	20	2.0	23
M2	1.7	19	2.5	28	3.6	41	4.3	49
M2.5	3.4	38	5.1	58	7.5	85	8.8	99
M3	5.8	66	8.9	100	13.0	147	15.2	172
M3.5	9.0	102	13.7	155	20.1	227	23.5	266
M4	13.4	151	20.4	231	30.0	339	35.0	396
M5	26.4	298	40.3	455	59.1	668	69.2	782

Table 7: Small Diameter Bolt Torque Values for M1.6 to M5

Thread Size	Tightening torque									
	Bolt Property Grade									
	8.8		10.9		12.9		HV (10.9)		V4A stainless steel AISI 316	
	Ft lbs	Nm	Ft lbs	Nm	Ft lbs	Nm	Ft lbs	Nm	Ft lbs	Nm
M8	18	25	27	36	31	42	--	--	--	--
M10	37	50	52	71	62	84	--	--	20	27
M12	64	87	91	124	107	145	74	100	37	50
M14	103	140	151	205	177	240	--	--	--	--
M16	160	220	230	310	270	360	185	250	85	115
M18	225	305	321	435	376	510	--	--	--	--
M20	310	420	440	600	520	700	332	450	--	--
M22	435	590	620	840	726	985	--	--	--	--
M24	540	730	760	1030	890	1210	590	800	--	--

Table 8: Small Diameter Bolt Torque Values for M8 to M24

Thread Size	Tightening torque							
	Bolt Property Grade							
	5.8		8.8		10.9		12.9	
	Ft lbs	Nm	Ft lbs	Nm	Ft lbs	Nm	Ft lbs	Nm
M6	3.8	5.2	5.8	7.9	8.6	11.6	10.0	13.6
M7	6.2	8.4	9.5	12.9	13.9	18.9	16.3	22.1

Table 9: Small Diameter Bolt Torque Values for M6 and M7

NOTICE

- ▶ Tightening torque values are based on VDI 2230, version 2003.
- ▶ Tightening torque utilization of yield strength 70 %.
- ▶ Total coefficient of friction between 0.15 to 0.21.

7.2.2 Standard Bolts - Small Diameter Bolt Torque Values

Size	Grade			
	5		8	
	Ft lbs	Nm	Ft lbs	Nm
#4x40	0	0	1	1.4
#5x40	1	1.4	1	1.4
#6x32	1	1.4	1	1.4
#8x32	1	1.4	2	2.7
#10x24	2	2.7	3	4.1
1/4x20	5	6.8	7	9.5
5/16x18	10	14	14	19
3/8x16	18	24	26	35
7/16x14	29	39	41	56
1/2x13	44	60	62	84
9/16x12	64	87	90	122
5/8x11	88	119	124	168
3/4x10	156	212	220	298
7/8x9	231	313	355	481
1x8	346	469	532	721
1-1/4x7	655	888	1063	1441
1-1/2x6	1140	1546	1849	2507

Size	Grade			
	5		8	
	Ft lbs	Nm	Ft lbs	Nm
#4x48	0	0	1	1.4
#5x44	1	1.4	1	1.4
#6x40	1	1.4	1	1.4
#8x36	2	2.7	2	2.7
#10x32	2	2.7	2	2.7
1/4x28	6	8.1	8	10.8
5/16x24	11	15	16	21.7
3/8x24	11	15	29	39
7/16x20	32	43	46	62
1/2x20	50	68	70	95
9/16x18	71	96	100	136
5/8x18	99	134	140	190
3/4x16	174	236	245	332
7/8x14	251	340	391	530
1x12	378	513	582	789
1-1/4x12	726	984	1177	1596
1-1/2x12	1285	1742	2084	2826

Table 10: Small Diameter Bolt Torque Values for Standard Bolts

7.2.3 Electrical Connection Torque Values

Screw or Bolt Size		Tightening Torque	
Metric	SAE	N-m	Ft-lbs
M5 or smaller	No. 8 or smaller	2	1.5
M5	No. 10	3	2.0
M6	1/4 in	8	6
M8	5/16 in	15	11
M10	3/8 in	26	19
-	7/16 in	41	30
M12	1/2 in	54	40
Larger than M12	9/16, 5/8 in or larger	75	55

Table 11: Electrical Connection Torque Values

7.2.4 Terminal Bolt Torque for Battery Connections (In Hub)

Battery Model	Tightening Torque	
	in-lbs	Nm
3EP & XE13	50	5.6
16EP & XE16	50	5.6
26EP & XE30	60	6.8
42EP & XE40	60	6.8
70EP & XE70	60	6.8

Table 12: Terminal Bolt Torque for Battery Connections (In Hub)

NOTICE

Tightening torque values are based on UL 486A-486B.

- ▶ For tower bus bar system torque values, refer to the GE Canada installation Manual 443, Manual 624 & Manual 672.
- ▶ For LVDP/converter electrical connections, refer to values listed in Section 7.2.3, taken from UL486A, unless vendor provided torque values are provided.

8 Torque Conversion Chart

Nm	Ft lbs	Nm	Ft lbs	Nm	Ft lbs
30	22	450	332	2500	1844
60	44	500	369	2700	1991
70	52	580	428	2800	2065
85	63	670	494	3080	2272
100	74	690	509	3500	2581
115	85	719	530	4350	3208
145	107	800	590	4500	3319
150	111	950	701	5000	3688
210	155	1700	1254	6000	4425
280	207	1780	1313	6101	4500
285	210	1850	1364		
300	221	1900	1401		
340	250	2000	1475		
350	258	2100	1549		

Table 13: Torque Conversion Chart

1 Nm = 0.7376 ft lbs

1.356 Nm = 1 ft lbs